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Primary treatment is the main treatment used to rid your body of lung cancer. The options for primary treatment depend heavily on the cancer stage.

Surgery is a common treatment for early and locally advanced non-small cell lung cancers (NSCLCs), but it's often not the only option. Read on to learn what your options for primary treatment may be.

Types of primary treatment

Many people with NSCLC have multiple cancer treatments. The treatment that is most important is called primary treatment.

The type of primary treatment differs among people with early or locally advanced NSCLC. Primary treatment may be:

- Surgery to remove tumors or body tissue with cancer
- Radiation therapy, which most often uses high-energy x-rays to treat lung cancer
- Chemoradiation, a combined treatment of prescribed drugs and radiation therapy

Your care team will recommend a primary treatment for you. Which primary treatment you receive will depend largely on the cancer stage. A cancer stage is a rating of the growth and spread of cancer.

"When you are deciding your treatment, remember that even though some decisions need to be made fast, don't rush. Think through your options and get second, or even third, opinions. Have people you can trust to talk through your options so you feel comfortable in your decisions."



Staging lung cancer

Lung cancer is staged using the tumor, node, metastasis (TNM) system maintained by the American Joint Committee on Cancer. Each letter in the system represents an area of cancer growth and is assigned a score.

T = Tumor score

The T score describes the primary tumor. The primary tumor is the main group of cancer cells in the lung.

The T scores are based on:

- The biggest width of the primary tumor measured in centimeters (cm)
- Invasive growth of the primary tumor into nearby body parts, such as the chest wall
- The number of tumors in a lung

The T scores used in this guide are T1, T2a, T2b, T3, and T4. Higher numbers represent more growth. A brief description of the T scores is in **Guide 5**.

Guide 5
T (tumor) scores used to stage lung cancer

T1	A T1 tumor is 3 cm or smaller. It's no larger than a grape.
T2a	A T2a tumor has one of these features: • Tumor is wider than 3 cm but no larger than 4 cm • Invasive growth into the next lobe, inner lining of the lung, or main airway • The tumor caused the lung to collapse or swell
T2b	A T2b tumor is based on its width. It is larger than 4 cm but no larger than 5 cm. For comparison, a golf ball is 4.3 cm. It may be invasive or affect the lung as described for a T2a tumor.
T3	A T3 tumor has one of these features: • Tumor is wider than than 5 cm but no larger than 7 cm • Invasive growth into the outer lining of the lung, heart lining, chest wall, azygos vein, a phrenic or chest spinal nerve, or stellate ganglion nerves • Multiple related tumors in the same lobe of the lung
T4	A T4 tumor has one of these features: • Tumor is wider than than 7 cm • Invasive growth into the heart, a major blood vessel, windpipe, esophagus, diaphragm, thymus, spine, vagus nerve, or a spinal nerve in the throat • Related tumors in more than one lobe of the lung

N = Node score

The N score describes cancer growth in nearby lymph nodes. Lymph nodes are small, bean-shaped structures located throughout your body that help fight disease. The N score is based on the spread of lung cancer to:

- Lymph nodes in the lung
- Lymph nodes just outside of the lung
- Lymph nodes far from the lung

The N scores used in this guide are N0, N1, N2a, N2b, and N3. Higher numbers mean greater spread of the cancer. A brief description of the N scores is in **Guide 6**.

M = Metastasis score

The M score tells you if the cancer has spread to the lining around the lung or to other organs. The spread of cancer is called metastasis.

Lung cancer tends to travel to the brain, liver, adrenal glands, and bones and from one lung to the other.

M0 means the cancer has not spread far. M1 means the cancer has spread far. There are three M1 scores to describe where the cancer has spread:

- M1a means the cancer has spread far within the chest.
- M1b means the cancer has spread to one place beyond the chest.
- M1c means the cancer has spread to more than one place beyond the chest.

Guide 6
N (node) scores used to stage lung cancer

N0	The cancer has not spread to lymph nodes.
N1	The cancer has grown into or spread to lymph nodes in the lung.
N2a	The cancer has spread to 1 lymph node in either of these places: • The middle of the chest next to the lung with cancer • Below the windpipe
N2b	The cancer has spread to 2 or more lymph nodes in one or both of these places: • The middle of the chest next to the lung with cancer • Below the windpipe
N3	The cancer has spread to lymph nodes in one or more of these places: • The middle of the chest near the other lung • The other lung • Near the collarbone

Lung cancer stages

There are 4 main stages of lung cancer. The cancer stages are further divided into subgroups and labeled with a letter.

- Stage 1 consists of stages 1A and 1B
- Stage 2 consists of stages 2A and 2B
- Stage 3 consists of stages 3A, 3B, and 3C
- Stage 4 consists of stages 4A and 4B

The TNM scores are combined to assign a stage to the lung cancer. TNM scores with a better outlook have a lower cancer stage. Read which TNM scores correspond with a cancer stage in **Guide 7**.

All lung cancers are staged before treatment based on the tests described in prior chapters. The pretreatment stage is called the clinical stage.

If you have surgery, the cancer is staged again. The second stage is called the pathologic stage. It's based on tests of tissue removed from your body.

Guide 7
Stages of lung cancer with TNM scores

Stage 1A	T1, N0, M0
Stage 1B	T2a, N0, M0
Stage 2A	T2b, N0, M0 T1, N1, M0
Stage 2B	T3, N0, M0 T1, N2a, M0 T2, N1, M0
Stage 3A	T3, N1, M0 T4, N0, M0 T4, N1, M0 T1, N2b, M0 T2, N2a, M0 T3, N2a, M0
Stage 3B	T2, N2b, M0 T3, N2b, M0 T4, N2a, M0 T4, N2b, M0 T1, N3, M0 T2, N3, M0
Stage 3C	T3, N3, M0 T4, N3, M0
Stage 4A	Any T, any N, M1a Any T, any N, M1b
Stage 4B	Any T, any N, M1c

Treatment by clinical stage

The clinical stage is very important for planning primary treatment. It helps predict the course of the cancer and which treatment is best. Options for primary treatment by clinical stage are noted in **Guide 8**.

When possible, surgery is the primary treatment of NSCLC. However, surgery isn't an option if it's unlikely that all the cancer can be removed.

Instead of surgery, radiation therapy or chemoradiation may be the primary treatment. When the goal is to cure cancer, these treatments are called definitive radiation therapy and definitive chemoradiation. Radiation therapy is the option for stage 1 and stage 2 cancers that haven't grown outside of

Guide 8
Options for primary treatment of early and locally advanced NSCLC

Clinical stage	TNM score	Surgery	Definitive radiation therapy	Definitive chemoradiation
1A	T1, N0, M0	●	●	
1B	T2a, N0, M0	●	●	
2A	T2b, N0, M0	●	●	
2A	T1, N1, M0	●		●
2B	T3 (no invasive growth), N0, M0	●	●	
2B	T3 (invasive growth), N0, M0 T1, N2a, M0 T2, N1, M0	●		●
3A	T3, N1, M0 T4, N0, M0 T4, N1, M0 T1, N2b, M0 T2, N2a, M0 T3, N2a, M0	●		●
3B	T2, N2b, M0 T3, N2b, M0	●		●
3B	T4, N2a, M0 T4, N2b, M0 T1, N3, M0 T2, N3, M0			●
3C	T3, N3, M0 T4, N3, M0			●

the lung and haven't spread to lymph nodes. Other cancers are treated with chemoradiation.

Your team will make a final recommendation for treatment based on additional factors. These factors include any challenges to treatment, how well your lungs are working, and your overall health. Your team's recommendation will also reflect your values and preferences.

"I tell newly diagnosed lung patients that today there is HOPE and better treatments and outcomes to the disease. In the past decade, the survivor rate has been climbing every year."

Participating in a clinical trial

A clinical trial may be an option for you.

A clinical trial is a type of medical research study. After being developed and tested in a lab, potential new ways of fighting cancer need to be studied in people.

If found to be safe and effective in a clinical trial, a drug, device, or treatment approach may be approved by the U.S. FDA.

Everyone with cancer should carefully consider all of the treatment options available for their cancer type, including standard treatments and clinical trials. Talk to your doctor about whether a clinical trial may make sense for you.

Phases

Most cancer clinical trials focus on treatment and are done in phases.

- **Phase 1** trials study the safety and side effects of an investigational drug or treatment approach.
- **Phase 2** trials study how well the drug or approach works against a specific type of cancer.
- **Phase 3** trials test the drug or approach against a standard treatment. If the results are good, it may be approved by the FDA.
- **Phase 4** trials study the safety and benefit of an FDA-approved treatment.

Who can enroll?

It depends on the clinical trial's rules, called eligibility criteria. The rules may be about age, cancer type and stage, treatment history, or

general health. They ensure that participants are alike in specific ways and that the trial is as safe as possible for the participants.

Informed consent

Clinical trials are managed by a research team. This group of experts will review the study with you in detail, including its purpose and the risks and benefits of joining. All of this information is also provided in an informed consent form. Read the form carefully and ask questions before signing it. Take time to discuss it with people you trust. Keep in mind that you can leave and seek treatment outside of the clinical trial at any time.

Will I get a placebo?

Placebos (inactive versions of real medicines) are almost never used alone in cancer clinical trials. It's common to receive either a placebo with a standard treatment, or a new drug with a standard treatment. You will be informed, verbally and in writing, if a placebo is part of a clinical trial before you enroll.

Are clinical trials free?

There is no fee to enroll in a clinical trial. The study sponsor pays for research-related costs, including the study drug. But you may need to pay for other services, like transportation or childcare, due to extra appointments. During the trial, you will continue to receive standard cancer care. This care is often covered by insurance.



Finding a clinical trial

In the United States

NCCN Cancer Centers
NCCN.org/cancercenters

The National Cancer Institute (NCI)
cancer.gov/about-cancer/treatment/clinical-trials/search

Worldwide

The U.S. National Library of Medicine (NLM)
clinicaltrials.gov

Need help finding a clinical trial?

NCI's Cancer Information Service (CIS)
1.800.4.CANCER (1.800.422.6237)
cancer.gov/contact

Key points

- The primary treatments of early and locally advanced non-small cell lung cancers (NSCLCs) are surgery, radiation therapy, and chemoradiation.
- Your treatment options are based on a rating of the growth and spread of the cancer before treatment. This rating is called the clinical stage. There are 4 stages.
- Surgery is a standard treatment for NSCLC. If surgery is not an option, radiation therapy or chemoradiation may be used for treatment instead.
- A clinical trial may be an option for you. Clinical trials test new ways of stopping cancer in people.

Questions to ask

- What is the clinical stage of the cancer?
- How did you decide what primary treatment I can have?
- Is there a clinical trial that's a good fit for me?

What's next?

Learn more about your primary treatment in the next chapters:

- Surgical procedures and treatments used with surgery are explained in *Chapter 6: Treating lung cancer with surgery*.
- The types of radiation therapy that are used to cure lung cancer are described in *Chapter 7: Treating lung cancer with radiation therapy*.
- Information on chemoradiation is in *Chapter 8: Treating lung cancer with chemoradiation.*

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Treating lung cancer with surgery

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